# SUPPORTS PROPER OVER THE BASE PULL BACK ALONG BASE CHANGE

THE STACKS PROJECT AUTHORS

*Editorial note.* Counted span is the author's own proof environment, source lines 1299-1307, immediately following the statement at 1283-1297.

*Editorial note.* The source writes (g')* for the pullback in the well-definedness line; the support identity is the span's own step.

*Allowed standard prerequisites (editorial).*

- Pullback of finite type modules. Modules on Sites, Lemma sites-modules-lemma-local-pullback, cited at proof lines 1300-1301. Makes the conclusion meaningful: (g')*F is finite type.
- Support of a pullback. Morphisms of Spaces, Lemma spaces-morphisms-lemma-support-finite-type, cited at proof lines 1302-1304. Supp((g')*F) = |g'|^{-1}(Supp(F)).
- Base change for proper-over-base closed subsets. Lemma lemma-base-change-closed-proper-over-base of this chapter, cited at proof lines 1305-1307. Terminal input: the closed-subset version proved earlier in this chapter.

*Source reference.* Original lemma 0CZF: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1116–1116. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 03DO: focused-seeds/geometry/S9-perf-base-change-module-support-proper-over-base/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 07TZ: focused-seeds/geometry/S9-perf-base-change-module-support-proper-over-base/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Lemma 0.1.** *Let $S$ be a scheme. Consider a cartesian diagram of algebraic spaces over $S$*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y \end{array}$$

*Date*: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*with $f$ locally of finite type. Let $\mathcal{F}$ be a finite type, quasi-coherent $\mathcal{O}_X$-module. If the support of $\mathcal{F}$ is proper over $Y$, then the support of $(g')^*\mathcal{F}$ is proper over $Y'$.*

*Proof.* Observe that the statement makes sense because $(g') * \mathcal{F}$ is of finite type by Modules on Sites, Lemma 03DO. We have $\mathrm{Supp}((g')^*\mathcal{F}) = |g'|^{-1}(\mathrm{Supp}(\mathcal{F}))$ by Morphisms of Spaces, Lemma 07TZ. Thus the lemma follows from Lemma 0CZF. □